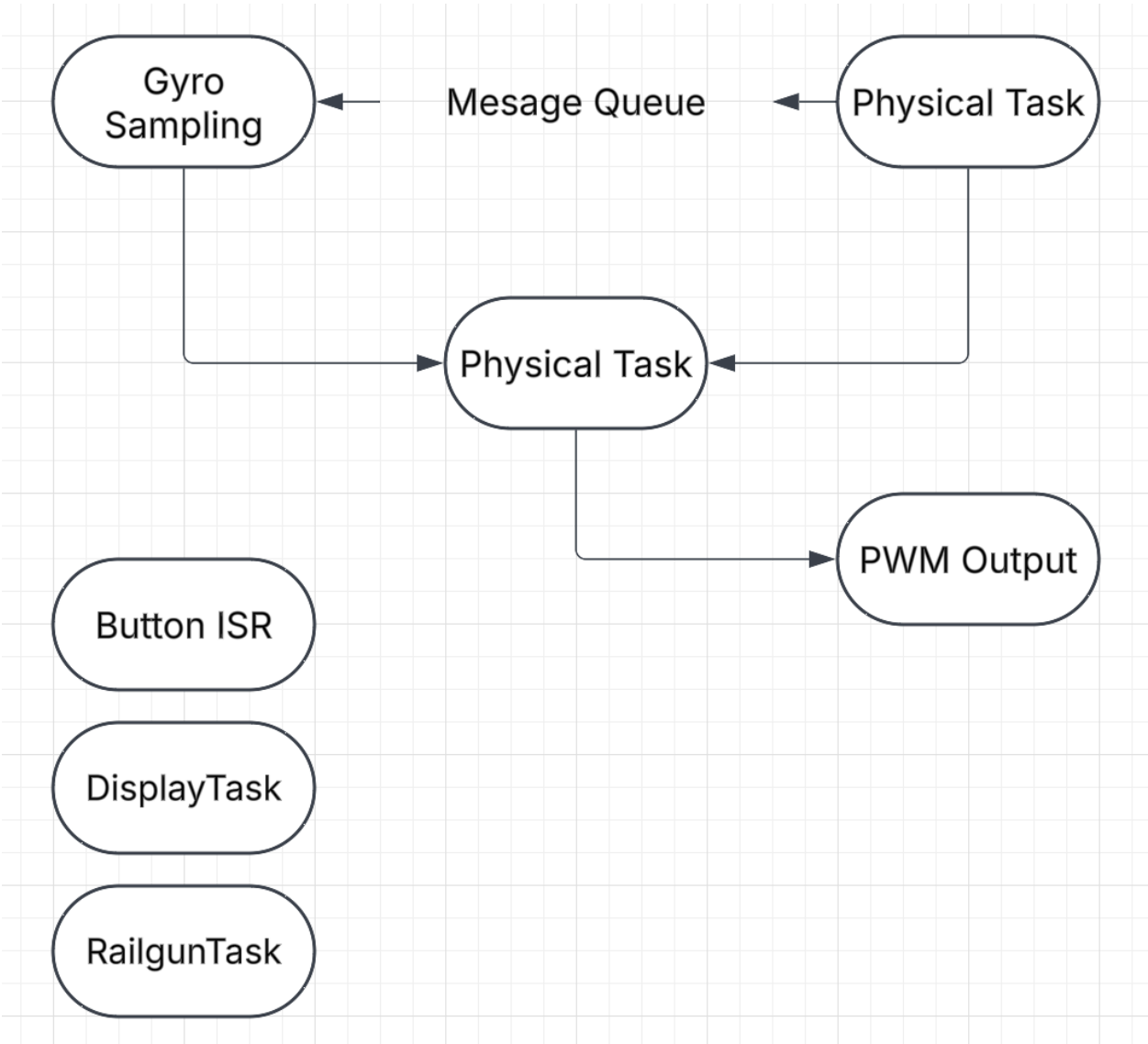
Joel J. Ramos Silva
ECEN 3753

Week 2 Project Report

Step 1: Task Diagram



Step 2: Test Plan and Results

UT1

- **Cut**: Before and after the DetectDoubleTap() logic.
- **Description**: Simulate fast consecutive button presses with known intervals.
- **Purpose**: Isolate detection timing boundaries and ensure only taps <300 ms apart register.

Joel J. Ramos Silva
ECEN 3753

UT2

- **Cut**: Around the capacitor energy drain loop in ShieldTask.
- **Description**: Test how energy is decremented every 2s using mocked time values.
- **Purpose**: Validate that energy depletes correctly and terminates the task at 0 energy.

Functional Tests

Test ID	Description	Status
FT1	Button press starts energy charging	Pass
FT2	Button release fires railgun	Pass
FT3	Double-tap activates shield	Pass
FT4	Shield drains energy every 2 seconds	Pass
FT5	Shield deactivates when energy reaches 0	Pass
FT6	Energy bar updates when firing railgun	Pass
FT7	Energy bar updates during shield drain	Pass
FT8	Cannot fire railgun with 0 energy	Pass
FT9	Cannot activate shield with 0 energy	Pass
FT10	Display updates show correct bar length for energy levels	Pass

Step 3: Project Status And Summary

Functionality + Usability

This week, I implemented both the railgun charging/firing logic and the shield activation system with real-time visual feedback on the LCD. Press-and-hold logic now tracks the duration, converts it into shot power, and fires a vertical beam. The shield logic is based on double-tap detection, activates a visual indicator, and drains energy every 2 seconds or 200ms. All elements are working on hardware, and feedback is immediate and clear.

Joel J. Ramos Silva
ECEN 3753

Effort & Estimate Summary

I have completed **47%** of my estimated work (**16.75 hrs completed / 36 hrs total estimate**) in **67%** of the project time (**24 hrs spent / 36 hr budget**).
For work that has been completed, I took **1.5x** as long as estimated (**24 / 16.75**).

Step 4: Task Table and Competition Comments

Task Name	**Est. Time (hrs)**	**Actual Time (hrs)**	**Status**
Task Diagram	1.5	1.5	Complete
Button Handling	3	3.5	Complete
Railgun Logic	4	6.0	Complete
Shield Logic	4	6.5	Complete
Display Task Setup	2	2.5	Complete
Cap Energy Visual Bar	2	2.0	Complete
Test Plan + UTs	2.5	2.0	Complete
Risk Register	1.5	1.0	Complete
Functional Test Summary	1.5	1.0	Complete
Projectile Logic	5	0.0	Not started
Satchel Explosion Logic	3.5	0.0	Not started

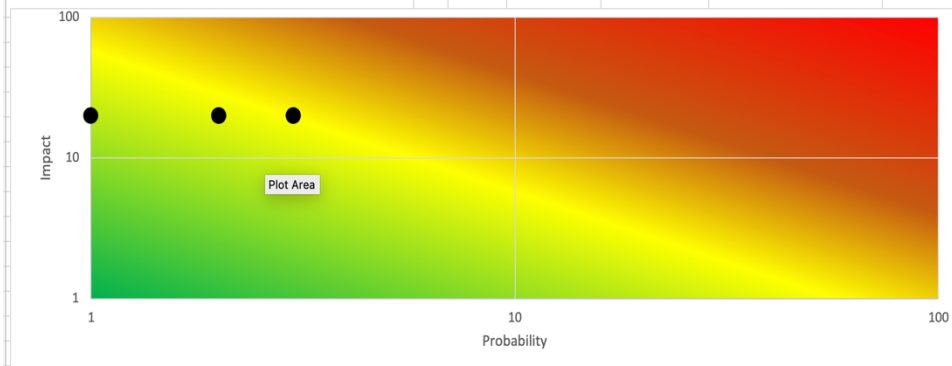
Total Estimate Completed: 16.75 hrs
Total Actual Time Spent So Far: 24 hrs

Completion Notes

- **Railgun Logic**: Learned how to convert press duration into beam strength to fire a visual beam.
- **Shield Logic**: First time implementing a double-tap ISR filter. Needed to fine-tune the debounce and timing thresholds. The shield activation and energy drain cycle is solid now.
- **Energy Bar Visuals**: Used vertical _ stacking to show capacitor energy and separate bars for railgun charge and shield health.

Joel J. Ramos Silva
ECEN 3753

- **Display Task**: Rearranged drawing logic into clear bar sections and added a clean rectangle-clear function to refresh the display without flicker.

Step 5: Risk Register

Item	P	I	Risk (P*I)	Recognized	Mitigated/ Resolved	ROAM	How
Shield stays on forever	3	20	60	26-Mar-25	Mitigated	M	Auto shuts off at 0 energy
Railgun fires with 0 energy	3	20	60	26-Mar-25	Mitigated	M	Condition added
Bar updates too slow	2	20	40	27-Mar-25	Resolved	A	Minor lag
Shield reactivation too quick	2	20	40	28-Mar-25	Resolved	R	enforced cool down
Incorrect task priority causes lag	1	20	20	29-Mar-25	Resolved	A	not critical now
			0				
			0				
			0				
			0